Answers learning task 9

2026 – 2027

Exercise 5.55: X and Y are independent, normal random variables with $E(X) = 2, V(X) = 5, E(Y) = 6$, and $V(Y) = 9$. Determine the following:

(a) $E(3X + 2Y)$
(b) $V(3X + 2Y)$
(c) $P(3X + 2Y < 18)$
(d) $P(3X + 2Y < 28)$

Answer

(a) 18
(b) 81
(c) 0.5
(d) 0.8667

Exercise 5.57: A plastic casing for a magnetic disk is composed of two halves. The thickness of each half is normally distributed with a mean of 3 millimeters, and a standard deviation of 0.1 millimeter and the halves are independent.

(a) Determine the mean and standard deviation of the total thickness of the two halves.

(b) What is the probability that the total thickness exceeds 6.3 millimeters?

Answer

(a) $\mu_{\text{sum}} = 6, \quad \sigma_{\text{sum}} = 0.1414$

(b) 0.0169

Exercise 5.60: The width of a casing for a door is normally distributed with a mean of 61 cm and a standard deviation of 0.4 cm. The width of a door is normally distributed with a mean of 60.5 cm and a standard deviation of 0.2 cm. Assume independence.

(a) Determine the mean and standard deviation of the difference between the width of the casing and the width of the door.

(b) What is the probability that the width of the casing minus the width of the door exceeds 0.8 cm?

Answer

(a) $\mu_{\text{diff}} = 0.5, \quad \sigma_{\text{diff}} = 0.4472$

(b) 0.2512

Exercise 5.63: The photoresist thickness in semiconductor manufacturing has a mean of 10 micrometers and a standard deviation of 1 micrometer. Assume that the thickness is normally distributed and that the thicknesses of different wafers are independent.

(a) Determine the probability that the average thickness of 10 wafers is either greater than

12 or less than 8 micrometers.

(b) Determine the number of wafers that need to be measured such that the probability that the average thickness exceeds 11 micrometers is 0.05.

(c) If the mean thickness is 10 micrometers, what should the standard deviation of thickness equal so that the probability that the average of 10 wafers is either more than 11 or less than 9 micrometers is 0.001?

Answer

(a) 0.0000

(b) 3

(c) 0.9610

Exercise 5.64: Assume that the weights of individuals are independent and normally distributed with a mean of 75 kg and a standard deviation of 15 kg. Suppose that 25 people squeeze into an elevator that is designed to hold 1950 kg.

(a) What is the probability that the load (total weight) exceeds the design limit?

(b) What design limit is exceeded by 25 occupants with probability 0.0002?

Answer

(a) 0.1587

(b) 2140.5063

Exercise 5.90: Suppose that $f_{XYZ}(x, y, z)$ is constant over the region $x^2 + y^2 \leq 1, 0 < z < 4$. Determine the following:

(a) $P(X^2 + Y^2 \leq 0.5)$ (b) $P(X^2 + Y^2 \leq 0.5, Z < 1)$
(c) Joint conditional probability density function of X and Y given that $Z = 1$
(d) Marginal probability density function of X
(e) Conditional mean of Z given that $X = 0$ and $Y = 0$
(f) Conditional mean of Z given that $X = x$ and $Y = y$

Answer

(a) 0.5 (b) 0.125
(c) $f_{XY|1}(x, y) = 1/\pi$ for $x^2 + y^2 \leq 1$
(d) $f_X(x) = \frac{2}{\pi}\sqrt{1 - x^2}$ for $-1 < x < 1$
(e) Since $f_{Z|0,0}(z) = 0.25$ for $0 < z < 4$, $\mu_{Z|0,0} = 2$
(f) Since $f_{Z|x,y}(z) = 0.25$ for $0 < z < 4$, $E(Z|(X, Y) = (x, y)) = 2$